Multivariate Calculus Lecture Notes
Section 12.6 – Quadric Surfaces

I. Cylindrical surfaces (show on Mathematica or Wolfram Alpha!)
   a. These are surfaces that are generated by taking a curve in two dimensions and extending it through the third dimension.
   b. Examples
      i. Elliptical cylinder (missing z variable): $x^2 + \frac{y^2}{4} = 1$

      ii. Parabolic cylinder (missing x variable): $y = z^2$

      iii. (Right) circular cylinder (missing y variable): $x^2 + z^2 = 16$

   c. Any two-dimensional graph can be extended to three dimensions in this manner.

II. Quadric surfaces
   a. The *quadric surfaces* are surfaces of the form $Ax^2 + By^2 + Cz^2 + Dxy + Exz + Fyz + Gx + Hy + Iz + J = 0$; i.e., surfaces with equations containing combinations of x, y, and z up to the second power.
   b. There are six basic types of quadric surfaces.
      i. Ellipsoid: $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$
         1. z constant: ellipse
         2. x constant: ellipse
         3. y constant: ellipse
      ii. Hyperboloid of one sheet: $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$
         1. z constant: ellipse
         2. x constant: hyperbola
         3. y constant: hyperbola
      iii. Hyperboloid of two sheets: $-\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$
         1. z constant
            - If $|z| > c$, then ellipse.

- If $|z| = c$, then point.
- If $|z| < c$, then nothing.

2. x constant: hyperbola
3. y constant: hyperbola

iv. Elliptical cone: $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} - \dfrac{z^2}{c^2} = 0$

1. z constant
   - If $|z| > 0$, then ellipse.
   - If $z = 0$, then point.
2. x constant
   - If $x = 0$ then straight lines.
   - If $x \neq 0$ then hyperbola.
3. y constant: same as with x

v. Elliptical paraboloid: $z = \dfrac{x^2}{a^2} + \dfrac{y^2}{b^2}$

1. z constant
   - If $z > 0$, then ellipse.
   - If $z = 0$, then point.
   - If $z < 0$, then nothing.
2. x constant: parabola
3. y constant: parabola

vi. Hyperbolic paraboloid: $z = -\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2}$

1. z constant: hyperbola
2. x constant: parabola
3. y constant: parabola

c. Examples

i. $y^2 = x^2 + z^2$

ii. $z = x^2 + y^2 + 1$

iii. $x^2 + y^2 - 4z^2 + 4x - 6y - 8z = -5$

1. Complete the square: $x^2-4x+4+y^2-6y+9-4(z^2-2z+1) = -5+4+9-4$
2. Simplify: $(x-2)^2 + (y-3)^2 - 4(z-1)^2 = 4$
3. Graphically:

III. Additional problems

a. Sketch the region bounded by the surfaces $z = \sqrt{x^2 + y^2}$, $x^2 + y^2 = 1$, and $z = 2$.

b. Sketch the region bounded by $z = x^2 + y^2$ and $z = 2 - x^2 - y^2$.

Homework: Section 12.6 #13, 15, 16, 18, 25, 26, 27, 29, 30, 38, 41, 42.